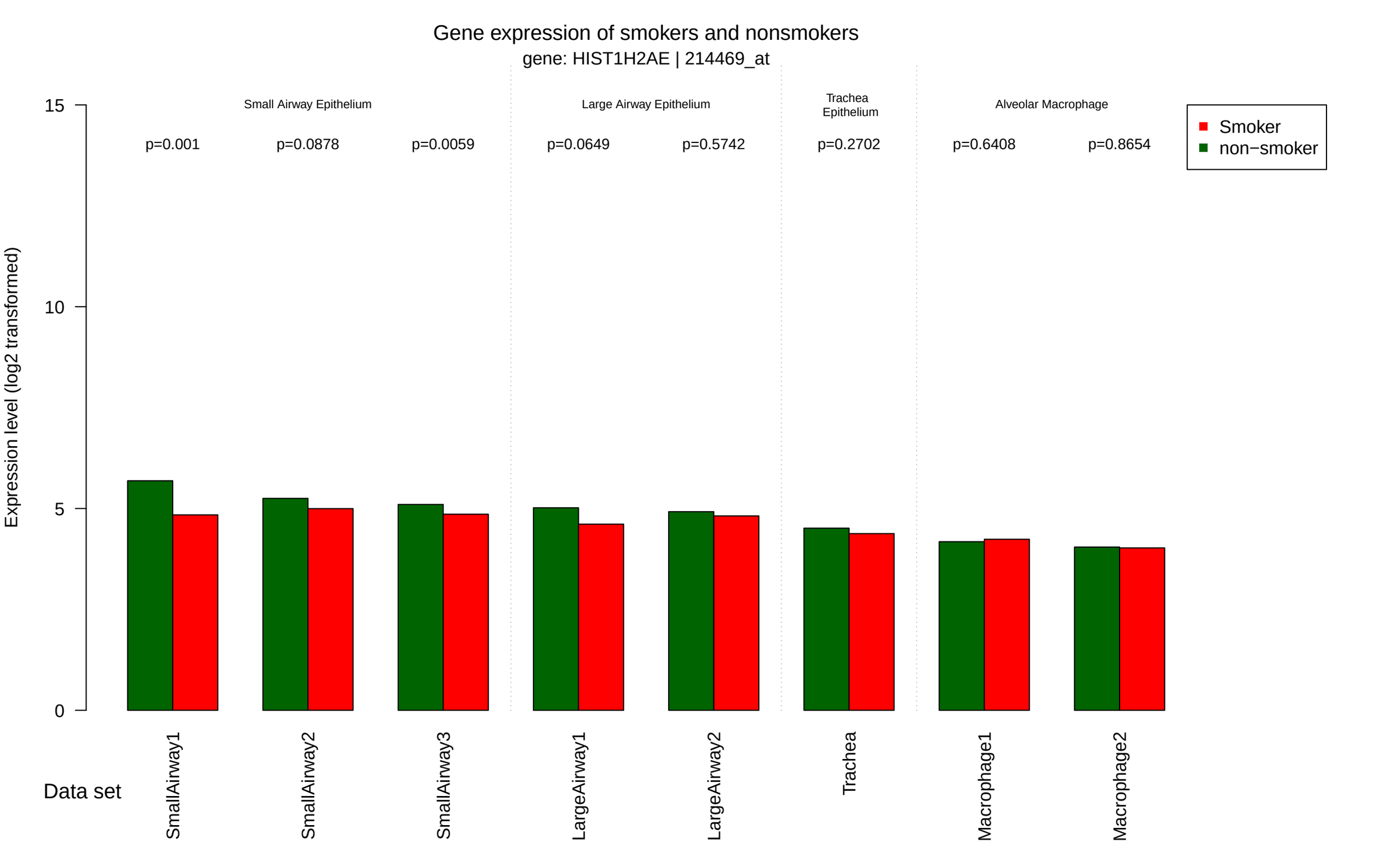
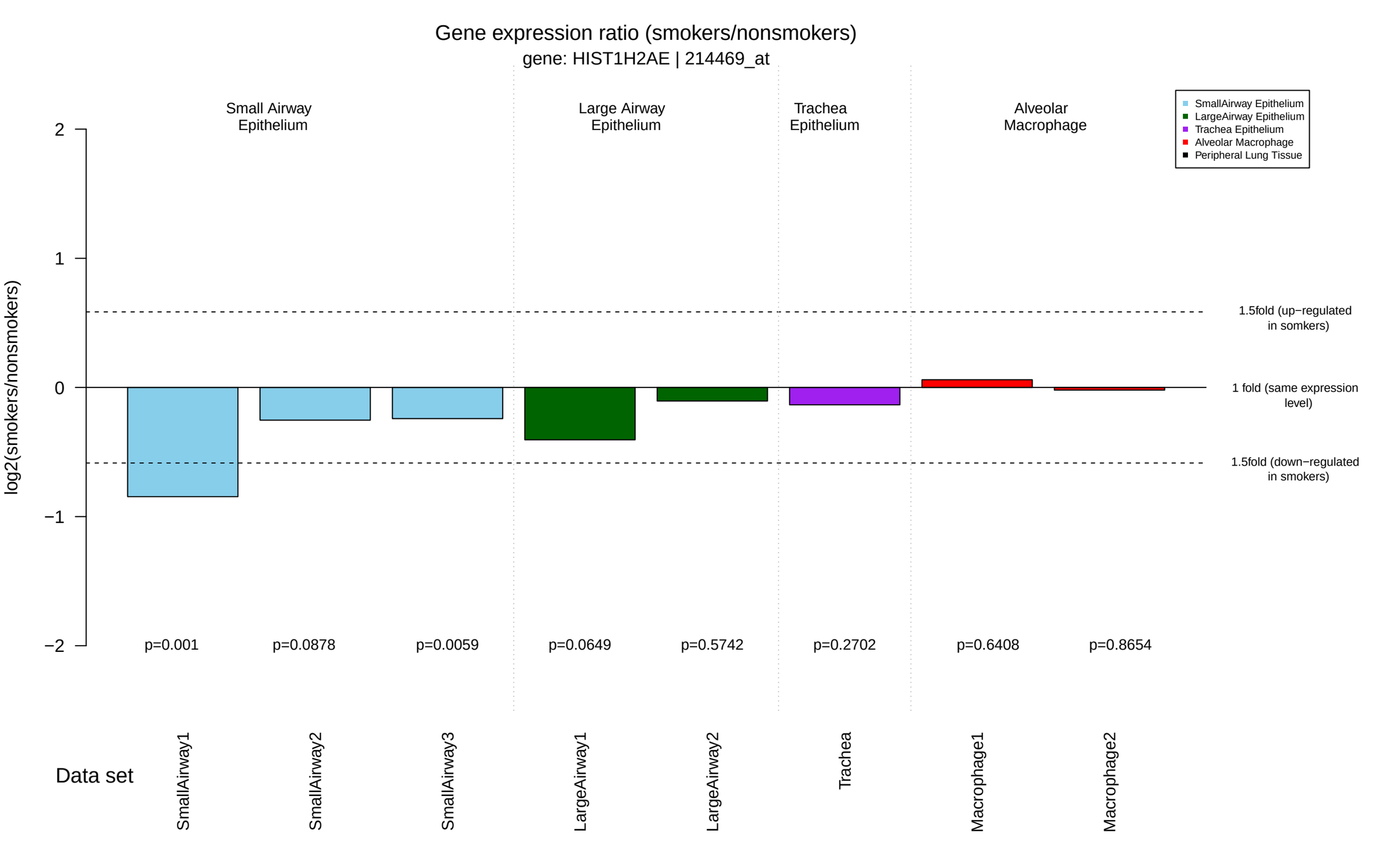
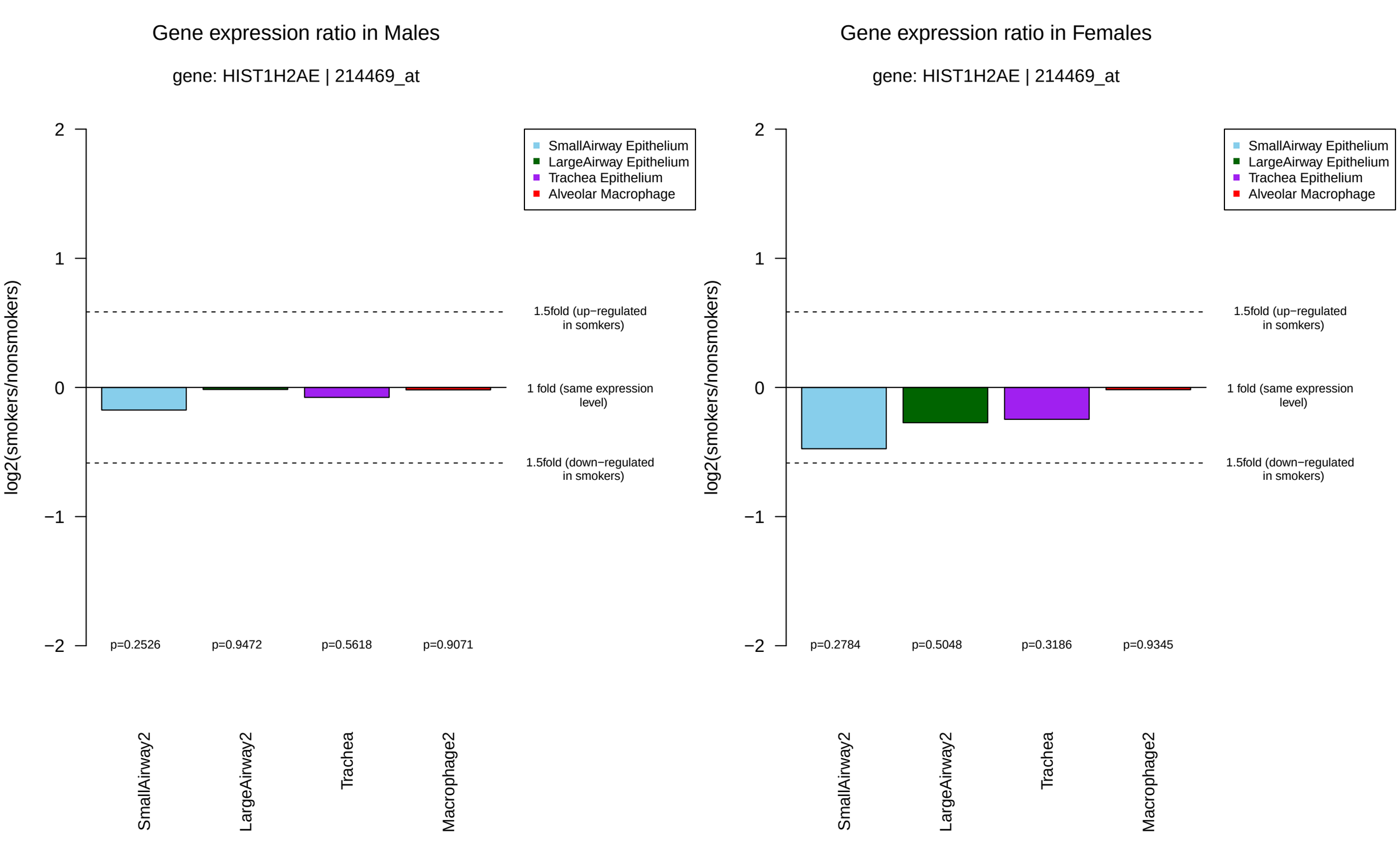
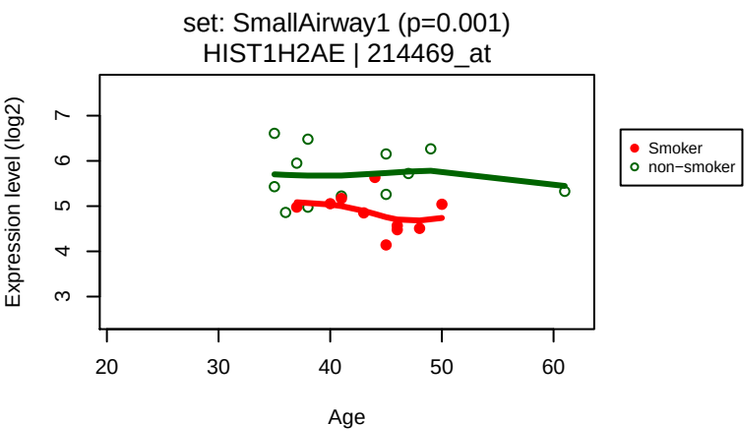
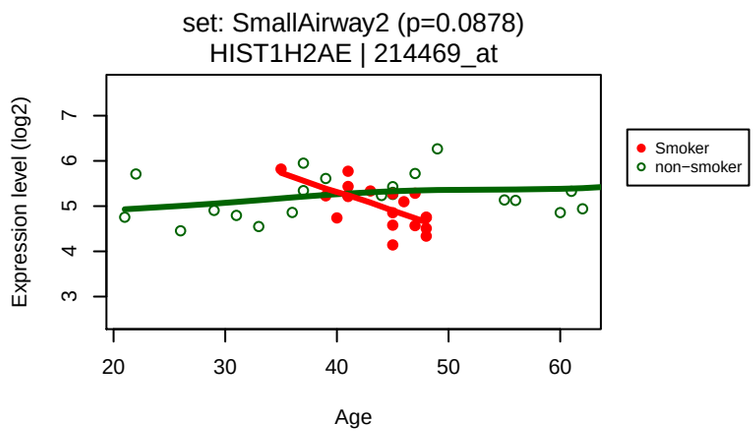
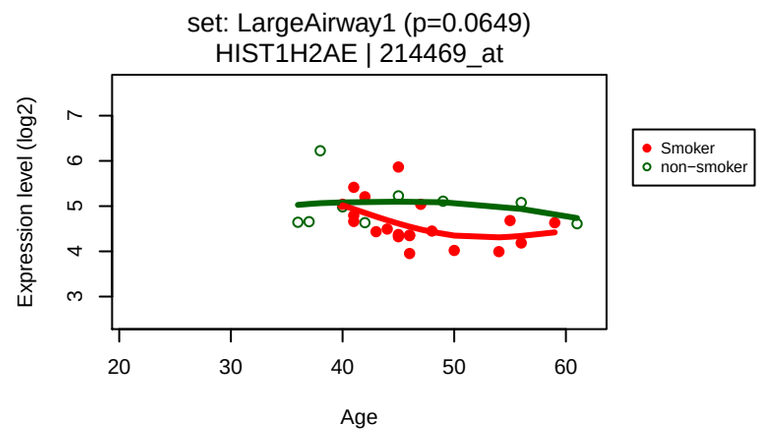

Gene expression of smokers and nonsmokers
gene: HIST1H2AE | 214469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.001
p=0.0878
p=0.0059
p=0.0649
p=0.5742
p=0.2702
p=0.6408
p=0.8654
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

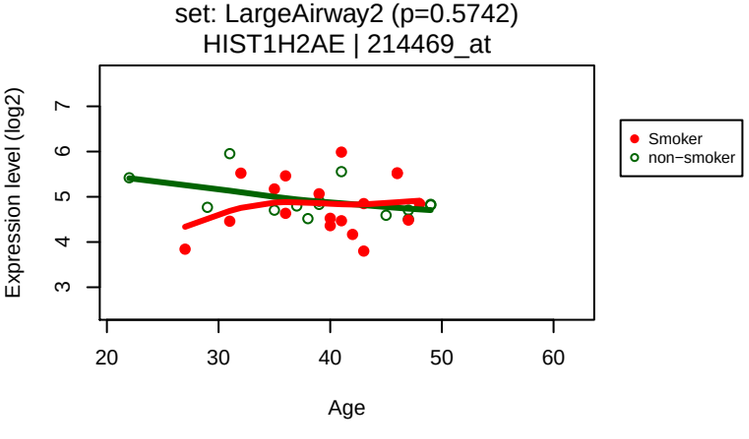

Gene expression ratio (smokers/nonsmokers)
gene: HIST1H2AE | 214469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.001
p=0.0878
p=0.0059
p=0.0649
p=0.5742
p=0.2702
p=0.6408
p=0.8654
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

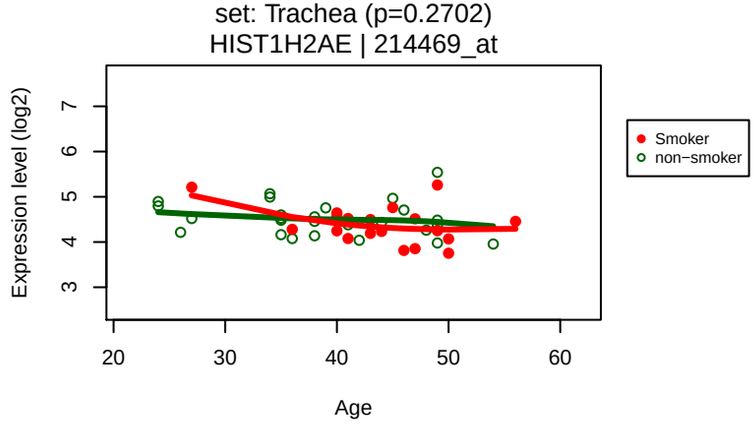

Gene expression ratio in Males
gene: HIST1H2AE | 214469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2526
p=0.9472
p=0.5618
p=0.9071
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIST1H2AE | 214469_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2784
p=0.5048
p=0.3186
p=0.9345
SmallAirway2
LargeAirway2
Trachea
Macrophage2

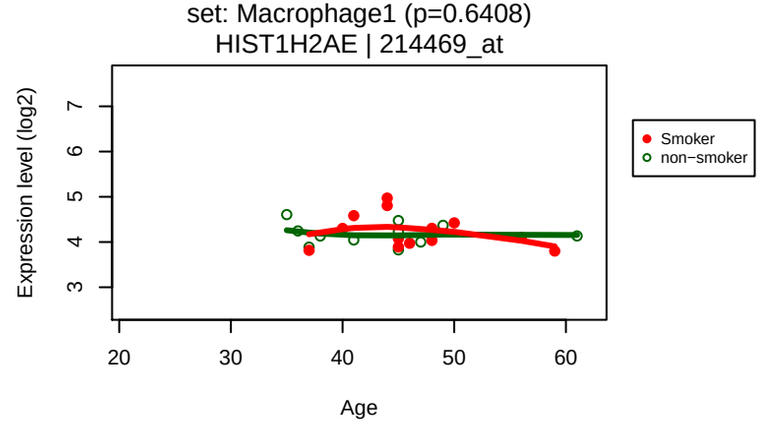

set: SmallAirway1 (p=0.001)
HIST1H2AE | 214469_at
Expression level (log2)
Age
Smoker
non−smoker

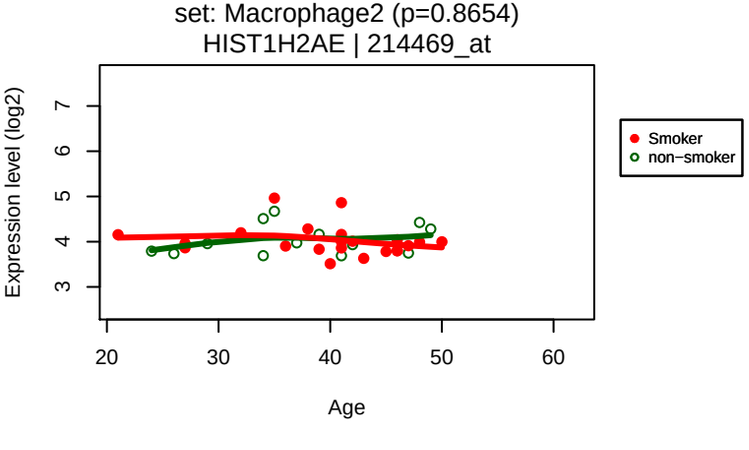

set: SmallAirway2 (p=0.0878)
HIST1H2AE | 214469_at
Expression level (log2)
Age
Smoker
non−smoker

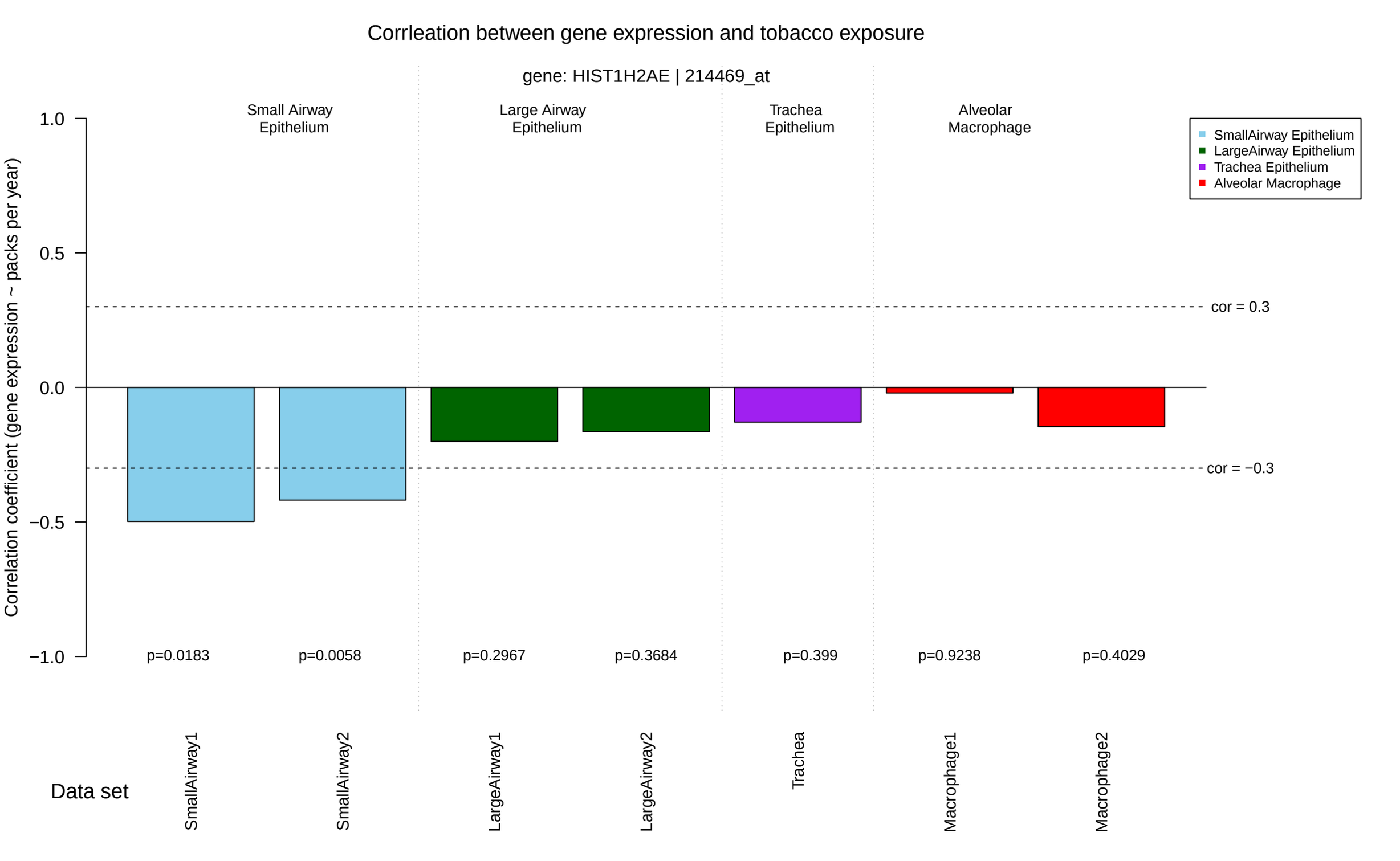

set: LargeAirway1 (p=0.0649)
HIST1H2AE | 214469_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5742)
HIST1H2AE | 214469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2702)
HIST1H2AE | 214469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6408)
HIST1H2AE | 214469_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8654)
HIST1H2AE | 214469_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIST1H2AE | 214469_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0183
p=0.0058
p=0.2967
p=0.3684
p=0.399
p=0.9238
p=0.4029
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2